Bariatric Meal Planner – AI Prompt Engineering Practice

Project Overview

This project is my **first AI-driven project** as I learn prompt engineering. The goal is **not the meal plan itself**, but to demonstrate my **ability to structure prompts, refine them, and analyze AI outputs**. This project shows:

- How I craft clear, specific prompts
- How I iterate and refine prompts to improve outputs
- How I document my AI workflow
- My early understanding of **applied AI in practical scenarios**

Project Context

- **Scenario:** I used the AI to generate a high-protein weekly meal plan for someone with dietary constraints.
- **Learning Objective:** Understand how prompt wording affects AI output, and practice refining prompts for clarity and structure.
- **Skills Demonstrated:** Prompt engineering, iterative testing, structured documentation, and workflow presentation.

1. Original Prompt

"My wife has had surgery to move part of her stomach to assist with losing weight. She now needs a high-protein diet but can only eat small amounts. Create a weekly lunch and dinner plan with quick purchase items and meals I can provide to her throughout the week."

Purpose:

- Test AI's ability to generate structured, actionable output
- Include constraints on portion size and protein content

2. Refined Prompt (Iteration)

"Imagine you are a nutrition assistant. Generate a 7-day high-protein weekly meal plan suitable for someone who can only eat small portions. Include lunch and dinner, focus on easy-to-prepare, grocery-store-available items, and provide clear tables with approximate protein grams per serving."

Notes:

- Added AI role for better context
- Requested a structured table format
- Goal: cleaner, more usable output

3. AI Output (Weekly Meal Plan)

Day	Lunch	Dinner	Protein (g)
Mon	Greek Yogurt + Nut Butter	Baked Chicken + Broccoli	20 / 25
Tue	Protein Shake + Banana	Scrambled Eggs + Spinach	22 / 24
Wed	Tuna Salad	Ground Turkey + Zucchini	24 / 25
Thu	Egg Salad	Salmon Fillet + Asparagus	20 / 26
Fri	Cottage Cheese + Peaches	Grilled Chicken + Peppers	20 / 25
Sat	Protein Smoothie	Baked Tilapia + Green Beans	22 / 26
Sun	Turkey Roll-Ups	Beef Stir-Fry + Peppers	23 / 25

4. Learning Notes

- Original prompt output was vague; refined prompt produced a **clear, structured table**
- Small changes in wording can drastically improve AI outputs
- Learned the importance of **AI context, specificity, and formatting instructions**
- Documented the process to show **iterative thinking and applied AI skills**